

Assignment of Chapter Two (A)

Fill-in-the-blank Questions:

- 1. . The Logic Algebra contains AND, OR, and NOT, such three basic functions.
 - 2. The three important rules in the logic algebra refer to Substitution rule, Dual rule and Complement rule.
3. In the logic algebra, if $A \cdot B = A + B$, then, $A=B$ is True.

Short Answer Questions: Prove the following logic functions

1. $A\bar{B} + \bar{A}C + B\bar{C} = \overline{ABC}(A + B + C)$

2. $AB\bar{C} + \overline{ABC} + \overline{AB} = \overline{ABC}$

3. $AB + \bar{A}C + \bar{B}C = AB + C$

4. $BC + D + \bar{D}(\bar{B} + \bar{C})(DA + B) = B + D$

5. $(A+B)(A+\bar{B})(\bar{A}+B)(\bar{A}+\bar{B}) = 0$

6. $AB + BC + AC = (A+B)(B+C)(A+C)$

7. $\overline{(\bar{A} + \bar{B} + \bar{C})(\bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C}) + \bar{A}\bar{B}\bar{C}} = (A+B+C)(AB+BC+AC) + ABC$

$$1. A\bar{B} + \bar{A}C + B\bar{C} = \overline{ABC}(A + B + C)$$

$$\text{Right hand side} = (\bar{A} + \bar{B} + \bar{C})(A + B + C)$$

$$= \bar{A}B + \bar{A}C + A\bar{B} + \bar{B}C + A\bar{C} + B\bar{C}$$

$$= \bar{A}B + \bar{A}C + A\bar{B} + A\bar{C} + B\bar{C}$$

$$= \bar{A}B + \bar{A}C + A\bar{B} + B\bar{C}$$

$$= \bar{A}C + A\bar{B} + B\bar{C} = \text{Left hand side}$$

$$\bar{B}C$$

$$A\bar{C}$$

$$\bar{A}B$$

Method 1

- 2. $AB\bar{C} + \overline{ABC} + \overline{AB} = \overline{ABC}$

Left hand side = $AB\bar{C} + \overline{AB(C + 1)}$

$$= AB\bar{C} + \overline{AB}$$

$$= \bar{C} + \overline{AB}$$

$$= \overline{ABC} = \text{right hand side}$$

Method 2

- 2. $AB\bar{C} + \overline{ABC} + \overline{AB} = \overline{ABC}$

Left hand side = $AB\bar{C} + (\bar{A} + \bar{B} + \bar{C})(\bar{A} + \bar{B})$

$$= AB\bar{C} + \bar{A} + \bar{A}\bar{B} + \bar{A}\bar{B} + \bar{B} + \bar{A}\bar{C} + \bar{B}\bar{C}$$

$$= AB\bar{C} + \bar{A} + \bar{B}$$

$$= AB\bar{C} + \overline{AB}$$

$$= \bar{C} + \overline{AB}$$

$$= \overline{ABC} = \text{Right hand side}$$

Method 3

- 2. $AB\bar{C} + \overline{ABC} + \overline{AB} = \overline{ABC}$

Left hand side = $AB\bar{C} + \overline{ABC} + \overline{AB}$

$$= AB\bar{C} + (\overline{AB} + \bar{C}) \overline{AB}$$

$$= AB\bar{C} + \overline{AB} + \overline{AB} \bar{C}$$

$$= AB\bar{C} + \overline{AB}$$

$$= \overline{AB} + \bar{C}$$

$$= \overline{ABC} = \text{right hand side}$$

Method 1

- 3. $AB + \bar{A}C + \bar{B}C = AB + C$

Left hand side = $AB + \bar{A}C + \bar{B}C$

$$= AB + \bar{A}C + BC + \bar{B}C$$

$$= AB + \bar{A}C + C(B + \bar{B})$$

$$= AB + \bar{A}C + C$$

$$= AB + C = \text{Right hand side}$$

Method 2

- 3. $AB + \bar{A}C + \bar{B}C = AB + C$

Left hand side = $AB + \bar{A}C + \bar{B}C$

$$= AB + (\bar{A} + \bar{B})C$$

$$= AB + \overline{ABC}$$

$$= AB + C = \text{Right hand side}$$

Method 3

- 3. $AB + \bar{A}C + \bar{B}C = AB + C$

Left hand side = $AB + \bar{A}C + \bar{B}C$

$$= AB + \bar{A}C + \bar{B}C + AC$$

$$= AB + \bar{A}C + AC + \bar{B}C$$

$$= AB + (\bar{A} + A)C + \bar{B}C$$

$$= AB + C = \text{Right hand side}$$

Method 1

- 4. $BC + D + \overline{D}(\overline{B} + \overline{C})(DA + B) = B + D$

$$\text{Left hand side} = BC + D + (\overline{B} + \overline{C})(DA + B)$$

$$= BC + D + \overline{BC}(DA + B)$$

$$= BC + D + DA + B$$

$$= B + D$$

Method 2

• 4. $BC + D + \overline{D}(\overline{B} + \overline{C})(\overline{D}A + B) = B + D$

Left hand side = $BC + D + (\overline{B} + \overline{C}) \overline{D} B$

$$= BC + D + (\overline{B} + \overline{C}) B$$

$$= \underline{BC} + D + \underline{B\overline{C}} = \underline{B(C + \overline{C})} + D$$

$$= B + D$$

- 5. $(A+B)(A+\bar{B})(\bar{A}+B)(\bar{A}+\bar{B})=0$

Left hand side $= A(\bar{A}+B)(\bar{A}+\bar{B}) = A\bar{A} = 0 = \text{Right hand side}$

Method 2:

Left hand side $= AB + A\bar{B} + \bar{A}B + \bar{A}\bar{B}$
 $= A + \bar{A} = 1 = \text{Right hand side}$

Method 1

- 6. $AB+BC+AC=(A+B)(B+C)(A+C)$

Right hand side= $(\cancel{AB}+AC+\underline{B}+\cancel{BC})(A+C)$

$$=(AC+\cancel{B})(A+C)$$

$$=\underline{AC}+\cancel{AC}+\underline{AB}+\underline{BC}=\underline{AC+AB+BC}=\text{Left hand side}$$

Method 2

- 6. $AB+BC+AC=(A+B)(B+C)\cancel{(A+C)}$

Right hand side= $(A+B)(AB+\cancel{BC}+AC+\underline{C})$

$$=(A+B)\cancel{(AB)}+C$$

$$=\cancel{AB}+AC+AB+BC=AB+BC+AC=\text{Left hand side}$$

- 7. $\overline{(\overline{A} + \overline{B} + \overline{C})(\overline{A}\overline{B} + \overline{B}\overline{C} + \overline{A}\overline{C}) + \overline{A}\overline{B}\overline{C}} = (A+B+C)(AB+BC+AC)+ABC$

Left hand side = $(\underline{ABC} + (A+B)\cancel{(B+C)}\cancel{(A+C)}) (\underline{A+B+C})$

$$= (ABC + \underline{AB+BC+AC}) (A+B+C)$$

$$= ABC(A+B+C) + (A+B+C)(AB+BC+AC)$$

$$= ABC + (A+B+C)(AB+BC+AC) = \text{Right hand side}$$

8. Find the **complement and dual forms** of the following functions

$$(1) \quad F = AB + \overline{A}\overline{B}$$

$$(2) \quad F = (A + B)(\overline{A} + C)(C + DE) + \overline{E}$$

$$(3) \quad F = [\overline{A}\overline{B} + (C + D)\overline{A}C]$$

$$(4) \quad F = A[\overline{B} + (C\overline{D} + \overline{E})G]$$

Solution:

$$(1) \quad F = AB + \overline{A}\overline{B}$$

Complement: $\overline{F} = \overline{AB + \overline{A}\overline{B}} = (\overline{A} + \overline{B})(A + B) = \overline{A}B + A\overline{B}$

Dual: $F' = (\overline{A} + \overline{B})(A + B) = \overline{A}B + A\overline{B}$

$$(2) \quad F = (A + B)(\overline{A} + C)(C + DE) + \overline{E}$$

$$\text{Complement : } \overline{F} = (\overline{A}\overline{B} + \overline{A}\overline{C} + \overline{C}(\overline{D} + \overline{E}))E = \overline{A}\overline{B}E + \overline{A}\overline{C}E + \overline{C}\overline{D}E$$

$$\text{Dual: } F' = (AB + \overline{A}\overline{C} + C(D + E))\overline{E} = AB\overline{E} + \overline{A}\overline{C}\overline{E} + CDE\overline{E}$$





$$(3) \quad F = \overline{[\overline{AB} + (C + D)\overline{AC}]}$$

Complement : $\overline{F} = \overline{AB} + \overline{AC} + \overline{AD} + D\overline{C}$

Dual:
$$\begin{aligned} F' &= A\overline{B} + (\overline{C} + \overline{D}) \bullet (A + C) \\ &= A\overline{B} + A\overline{C} + A\overline{D} + C\overline{D} \\ &= A\overline{B} + A\overline{C} + C\overline{D} \end{aligned}$$

$$(4) \quad F = A[\overline{B} + (C\overline{D} + \overline{E})G]$$

Complement : $\overline{F} = \overline{A} + \overline{B}\overline{C}\overline{E} + B\overline{D}\overline{E} + \overline{B}\overline{G}$

Dual:
$$F' = A + \overline{B}\overline{C}\overline{E} + \overline{B}\overline{D}\overline{E} + \overline{B}\overline{G}$$

